Medical appointment no-show analysis

SQL Server and SSMS for analysis. Excel for visualization.

Appointments	Missed appointments	No-show rate	Avg. lead time
110,521	**22,314**	**20.2%**	**10.2 days**
after data cleaning	NoShowFlag = 1	missed / appointments	calendar-day difference

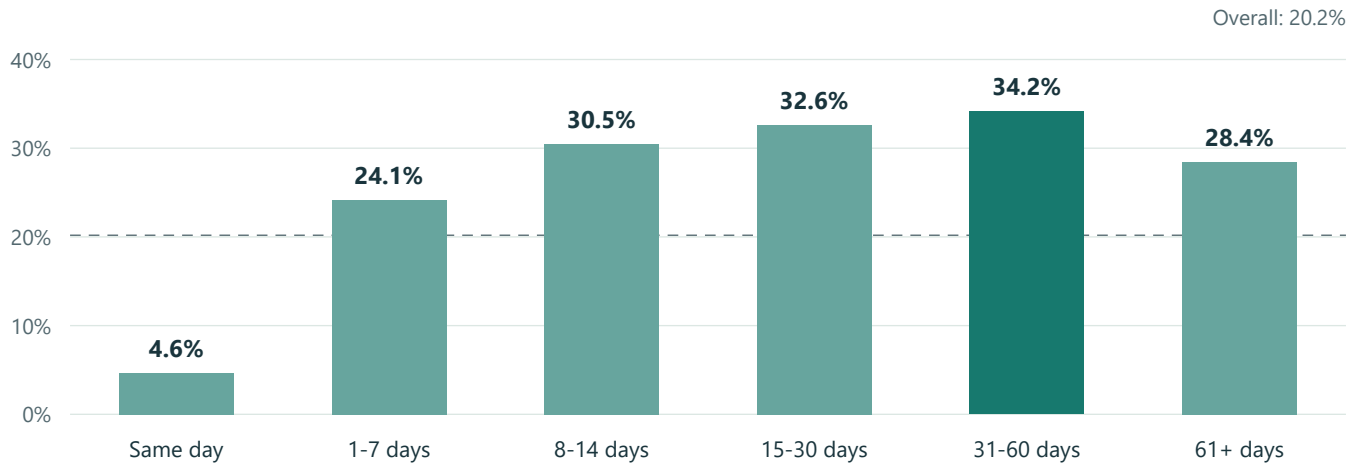
BUSINESS QUESTION

Where should appointment teams focus follow-up? The analysis compares no-show rates and missed counts to identify segments for a limited improvement pilot.

DATA AND APPROACH

110,527 source records, 14 columns; one row per appointment. A patient can appear more than once. Six invalid records were excluded from the analysis, with the Raw data preserved.
Workflow: Raw table > validation and typed Clean table > SQL view > Excel summaries and dashboard.

BOOKING LEAD TIME AND NO-SHOW RATE



Rate = missed / appointments within each band. Lead time is measured from booking to appointment.

HIGHEST RATE: 31-60 DAYS

34.2%: 2,829 missed from 8,283 appointments.

LARGEST MISSED COUNT: 1-7 DAYS

7,772 missed from 32,185 appointments (24.1%).

INTERPRETATION

Use **rate and volume together** when setting follow-up priorities. The relationship is not a steady increase: the 61+ day group falls to **28.4%**. These are observed associations, not proof that lead time causes absence.

Source: Joni Hoppen / Aquarela Analytics, Medical Appointment No Shows.
Kaggle dataset (CC BY-NC-SA 4.0). Historical appointment data: 29 Apr - 08 Jun 2016.

Findings to action

Prioritize a testable scheduling intervention, then measure the outcome.

ADDITIONAL FINDINGS

SEGMENT	EVIDENCE	USE IN DECISIONS
Age groups	18-34: **24.0%**; 65+: **15.5%**. Rates refer to appointments in each age group.	Investigate communication needs; check other group differences.
Neighbourhoods	Jardim Camburi: **1,465 missed, 19.0%**. Itararé: **923 missed, 26.3%**.	Separate follow-up volume from the share of appointments missed.
Daily trend	**27 observed dates** in a short historical period.	Monitor variation; annual seasonality is not established.

RECOMMENDED PILOT

01 TARGET Consider the 31-60 day segment for its high rate and the 1-7 day segment for its larger missed volume.

02 TEST Appointment management leads a limited pilot of confirmation, early cancellation and rescheduling; use a waitlist to refill available slots.

03 EVALUATE Use a suitable comparison group. Track no-show rate, slots actually refilled and follow-up workload; record reasons for absence.

WHAT-IF: 10% RELATIVE REDUCTION

Potential recovered
2,231
rounded appointment count

Remaining missed
20,083
total appointments unchanged

Scenario rate
18.2%
from a 20.2% baseline

An assumption, not an achieved improvement or validated forecast. Calculation: 22,314 x 10%, rounded to whole appointments; remaining no-shows divided by 110,521.

PREPARATION

Validate IDs, dates and field values; retain IDs as text. Exclude one negative-age record and five dates before booking. Map **1 = missed, 0 = attended**. Create lead-time and age groups for analysis.

SCOPE AND LIMITATIONS

Historical Brazilian data; local results may differ. Reasons for absence and intervention effects are not established. No measured financial savings or hospital-capacity gains are claimed.

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